

Introduction & Research

Why this course?

Need for formal, focused APPROACH.

Understanding of theories or techniques is necessary but not sufficient

➔ understanding of issues, literature, research process + ability to conduct research and communicate results to the stakeholders in an ethical manner.

What?

1. Overview of the conceptual and methodological basis of research

=> UNDERSTAND RESEARCH PROCESS

2. Methods for research: Statistics and Time Series

=> DEVELOP SKILLS IN RESEARCH

3. Procedural guidelines on planning, structuring, and conducting research projects

=> PREPARE RESEARCH PROPOSAL

(thesis, programs of TÜBİTAK or HorizonEurope, etc.)

3. Ethical principles

=> ETHICAL RESEARCH

How?

Week	Subjects
1	Introduction, Scientific research process
2	Identification of research problem, purpose and hypothesis
3	Research design, data collection, using libraries and resources, and AI
4	Review and Quiz-1
5-6	Statistical analysis
7-8-9-10	Methods for time series analysis and prediction
11	Research proposal
12-13	Ethics
14	Review and Quiz-2

Required Reading	Research Methodology, Ranjit Kumar, SAGE Publications, London, 2011. Lecture notes
Other Course Resources	Research Methods for Engineers, David V. Thiel, 2014 Research Methods: The Basics, Nicholas Walliman, Routledge Pub., 2011.

GRADING:

2 X QUIZES	30%
3 X HOMEWORK	40%
PROJECT	30%

KHAS LEARN course portal:
learn.khas.edu.tr)

Lectures on Wednesday btw. 12:00 and 14:50 @ Galata Hall
+ asynchronous self-study

Who?

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(+ double major degree in Mathematics)
- Research interests: IoT, computational intelligence
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Research Methodology vs Research Method

Research Methodology	Research Method
principles for organizing, planning, designing and conducting research	specific ways (procedures) to perform a task
• Reason for the research • Clear objectives • Complete reference to prior research • Understanding for the conceptual and theoretical basis of the research • Application of appropriate methods • Comparison and validation of results • Deriving conclusions from results	• Field dependent • Dependent on • Application • Objective • Mode of inquiry (Inquiry: investigation, looking into)

RESEARCH will include both

- Discovery-Exploration (finding and creating new knowledge)
- Confirmation (check validity or reliability of information)

Classification of Research?

Students?

Background?

Research experience?

Expectations?

Questions?

SGS 501/601 RESEARCH METHODS / RESEARCH METHODS and SCIENTIFIC ETHICS



The RESEARCH PROCESS



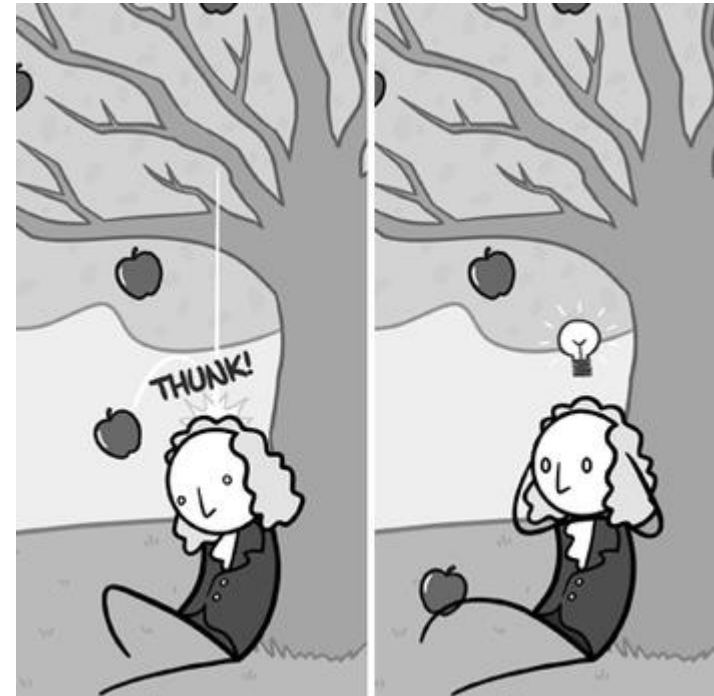
AGENDA

- What is research? And, what is not!
- Philosophy of Science
- Classification of Research
- Steps of Research Process

RESEARCH

RESEARCH is *the systematic approach to obtaining and confirming new and reliable knowledge.*

- Accidental discovery of new phenomenon is NOT research (may lead to a structured research process to verify or understand the observation)
- Data Collection is NOT research (an intermediate step of the research process to gain reliable knowledge)
- Searching and reviewing published research results is NOT research (an intermediate step of the research process to obtain information for analysis, synthesis and comparison)



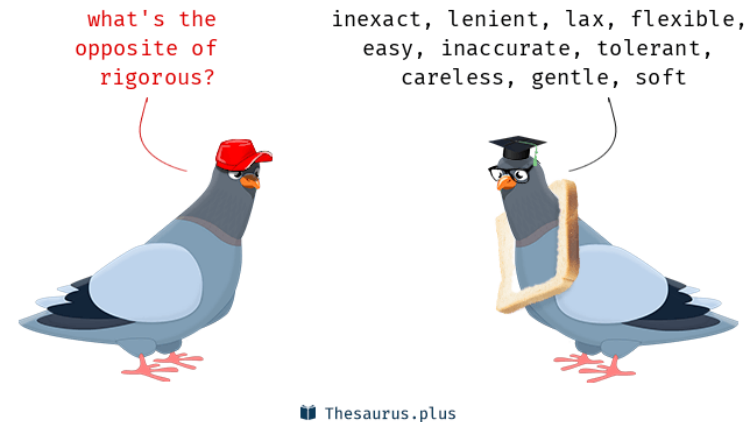
RESEARCH is

- Searching for explanation of events, phenomena, relationships and causes
 - What, how and why things occur
 - Are there interactions?
- A creative, circular process
 - Planned and managed – to make the information generated credible
 - **Circular** – always leads to more questions

Research usually leads to a benefit for humanity beyond the advancement of science.

CHARACTERISTICS of RESEARCH

- Controlled: effect $\leftrightarrow$ cause
- Rigorous (TR: titiz, sıkı): appropriate and justified procedures
- Systematic: sequence of procedures
- Valid and verifiable: correct and repeatable
- Empirical: evidence from measurements or observations
- Critical: pass critical investigation



CREATIVITY in RESEARCH

Research requires imagination, initiative, intuition, and curiosity.

- * Gather and use previously developed *knowledge*
- * Exchange ideas
- * Apply deductive logic
- * Look at things in alternate ways
- * Question or challenge assumptions
- * Search for patterns or relationships
- * Take risks
- * Cultivate tolerance for uncertainty
- * Allow curiosity to grow
- * Set problems aside ... and come back
- * Write down your thoughts
- * Freedom from distraction ...
some time to think.



Creativity may provide the difference between satisfactory and outstanding research.

SCIENCE and RESEARCH

Science is the organized accumulation of systematic [reliable] knowledge for the purpose of intelligent explanation/prediction

- Research and Science are mutually interdependent
 - Science is accumulated, reliable knowledge
 - Research is a process through which science is expanded and tested for validity

Does science consist of “factual truth”? Do values and personal views have a place in science?

PHILOSOPHY of SCIENCE-I:

Idealism vs Materialism

Two extremes in metaphysics (branch of philosophy dealing with being, knowing, identity, time, and space: first principles of things)

Idealism, that advocates that **reality is all in the mind**, that everything that exists is in some way dependent on the activity of the mind. Hence, as phenomena are reliant on mental and social factors they are therefore in a state of constant change e.g. music is not just sound, it is an emotional experience.

VS

Materialism (or reductionism), that insists that **only physical things and their interactions exist** and that our minds and consciousness are wholly due to the active operation of materials. Hence, **phenomena are** independent of social factors and are therefore **stable** e.g. music is just vibrations in the air.

(From: The Basics, Nicholas Walliman, Routledge Pub., 2011)

PHILOSOPHY of SCIENCE-II:

Induction vs Deduction

Epistemology (theory of knowledge, methods to obtain and validate knowledge):
empiricism – knowledge gained by sensory experience (using inductive reasoning);
rationalism – knowledge gained by reasoning (using deductive reasoning).

INDUCTION:

- Reasoning from the specific circumstances or outcomes to a conclusion about general circumstances or outcomes
- Empirical process of reaching a conclusion or arriving at new principles from known data and experience by observing objective realities.

Problem: need of large number of observations and repetition

DEDUCTION:

- The process of reasoning from general premises (e.g. assumptions) to specific results or conclusions.
- Establish a series of assumptions about conditions and behaviors, and logically work through to variables and parameters we wish to explain or predict.

Problem: need of falsifiable hypotheses. What if assumptions are wrong?



Examples of Induction and Deduction

INDUCTION:

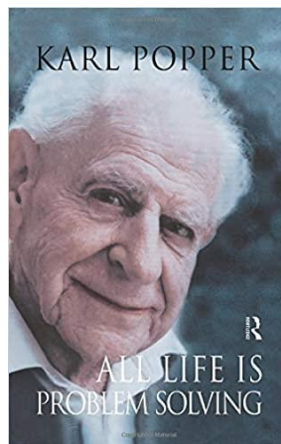
DEDUCTION:

PHILOSOPHY of SCIENCE-III:

SCIENTIFIC METHOD

The creative middle way: Hypothetico-Deductive Reasoning

- identification or clarification of a problem;
- developing a hypothesis (testable theory) inductively from observations;
- charting their implications by deduction;
- practical or theoretical testing of the hypothesis;
- rejecting or refining it in the light of the results.



“Whenever a theory appears to you as the only possible one, take this as a sign that you have neither understood the theory nor the problem which it was intended to solve.”

— **Karl Popper**

PHILOSOPHY of SCIENCE-IV

POSITIVISM vs RELATIVISM (INTERPRETIVISM, IDEALISM, CONSTRUCTIVISM, CONSTRUCTIONISM)

(From: The Basics, Nicholas Walliman, Routledge Pub., 2011)

POSITIVISM: the world around us is real, and that we can find out about these realities (materialism, reductionism)

Knowledge is derived using scientific method and based on sensory experience gained through experiments or comparative analysis.

RELATIVISM: view of the world that we see around us is the creation of the mind (idealism). This does not mean that the world is not real, but rather that we can only experience it personally through our perceptions which are influenced by our preconceptions, beliefs and values; we are not neutral, disembodied observers but part of society.



POSITIVISM vs RELATIVISM in RESEARCH

(From: The Basics, Nicholas Walliman, Routledge Pub., 2011)

Issue	Positivism	Relativism
Basis	Realism: the world exists and is knowable as it really is.	Idealism: the world exists but different people construe it in very different ways.
Research	To discover universal laws and generalizations	To reveal different interpretations of the world as made by people
Researcher	Neutral observer.	Part of the research process.
Approach	Rational, using inductive and scientific methods and value free data.	Subjective, using inductive methods and value laden data.
Methods	Experiments or mathematical models and quantitative analysis to validate, reject or refine hypotheses.	Surveys and observations with qualitative analysis to seek meaningful relationships and the consequences of their interactions. Analysis of language and meaning.
Society	Search for order.	Search for dynamics.

NON-SCIENTIFIC APPROACH

- Intuition, belief
 - Consensus
 - Authority
 - Casual Observation
 - Informal logic
- Opinion
- Biased/Flawed

From: Coursera: Quantitative RM, Annemarie Zand Scholten, U. of Amsterdam

SCIENTIFIC METHOD-I

- Empirically testable
“observations/measurements”
- Replicable “repeatedly done tests”
- Objective “independent of researcher”
- Transparent “publicly available, replicable by anyone”
- Falsifiable “a contradiction is possible”
- Logically consistent “internal consistency”

From: Coursera: Quantitative RM, Annemarie Zand Scholten, U. of Amsterdam

SCIENTIFIC METHOD-II

Interaction of

- Fact – a verifiable observation through sense or measurement => independent of personal judgment
- Hypothesis – a testable proposition, tentative assertion of a relationship between factors or events that is subject to verification
- Theory – abstractions, simplifications, and/or generalizations as logical relationships among facts established by deduction

[use induction to validate/evaluate a theory]

CLASSIFICATION of RESEARCH

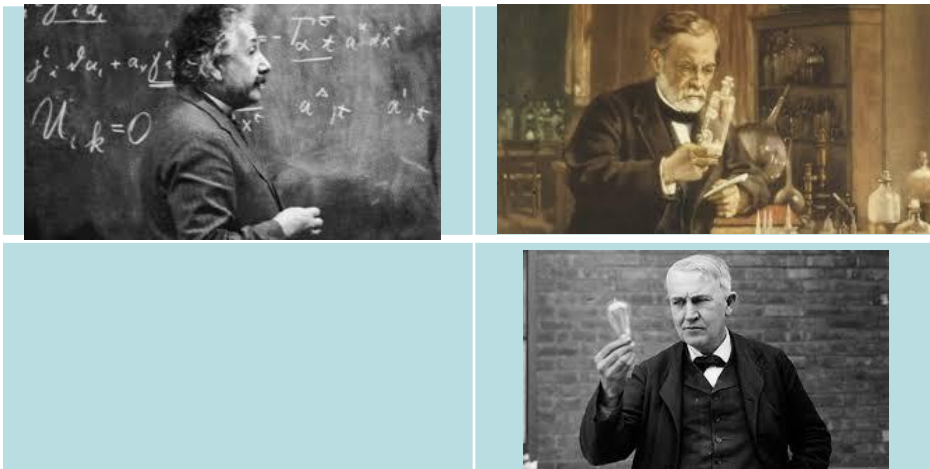
Based on

- APPLICATION: { Pure/Fundamental/Basic, Applied}
- OBJECTIVES: {Descriptive, Correlational, Exploratory, Explanatory}
- MODE of INQUIRY: {Quantitative, Qualitative}

RESEARCH by TYPE of APPLICATION

- PURE or FUNDAMENTAL or BASIC RESEARCH : establish fundamental facts and relationships within a discipline or field of study → leads to theories

Low ←- fundamental understanding → High



Low ←----- practical use -----> High



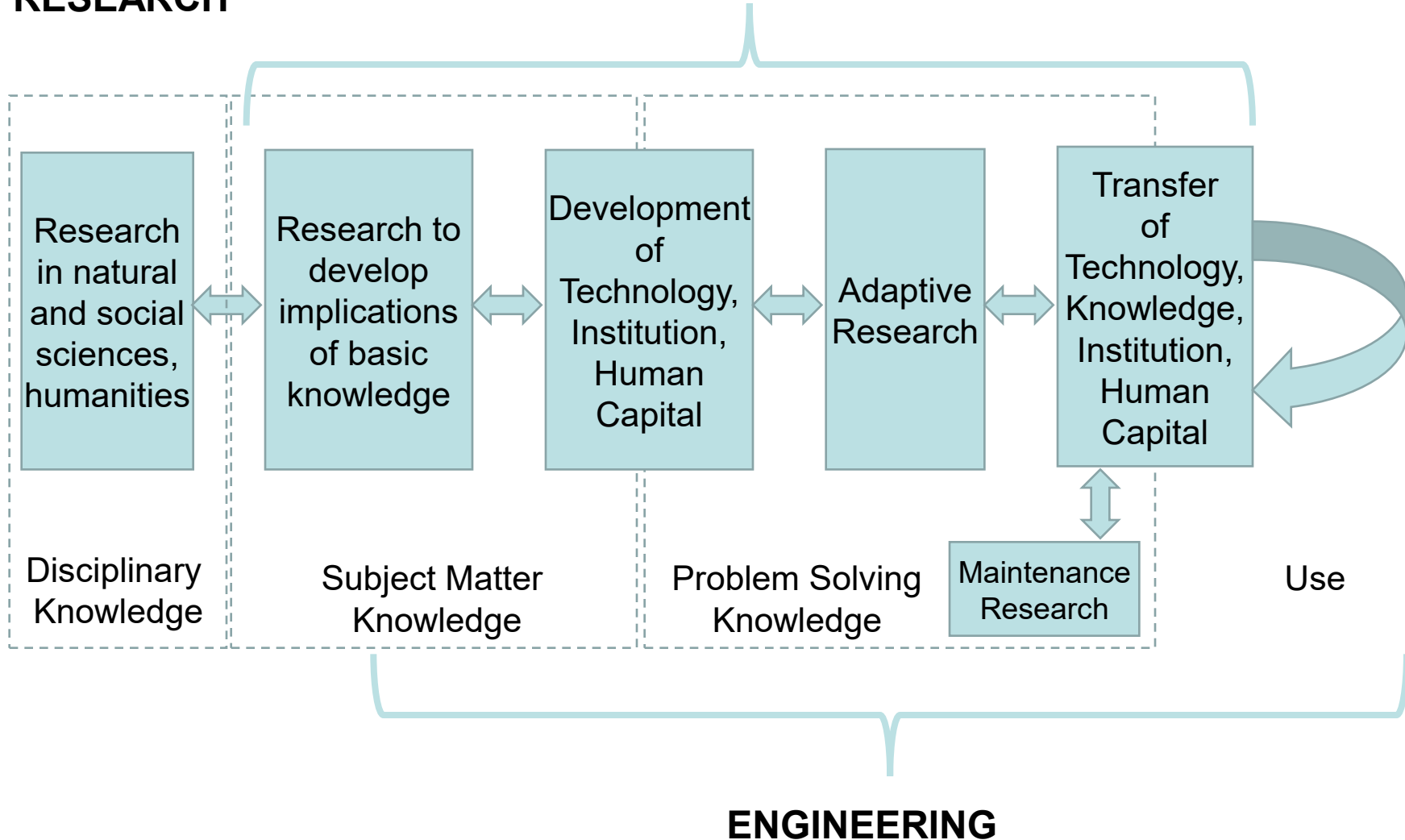
- APPLIED RESEARCH : obtaining information to help resolve a particular problem, may be guided by basic research → leads to practical solution for real life problem

RELATION btw BASIC and APPLIED RESEARCH

**BASIC
RESEARCH**

APPLIED RESEARCH

USE



RESEARCH by TYPE of OBJECTIVE

- Descriptive: attempt to determine, describe, or identify something, the intent is often synthesis, which pulls knowledge or information together
- Correlational: to discover or establish the existence of a relationship / association / interdependence between two or more aspects of a situation.
- Exploratory: to explore an area where little is known or to investigate the possibilities of undertaking a particular research study
- Explanatory/Analytic: attempt to establish why something occurs or how it came to be

RESEARCH by TYPE of INQUIRY MODE

- Qualitative

- purpose is primarily to describe a situation, phenomenon, problem or event;
- information is gathered through the use of variables measured on nominal or ordinal scales (qualitative measurement scales);
- analysis is done to establish the variation in the situation, phenomenon or problem without quantifying it.

- Quantitative

- purpose is to quantify the variation in a phenomenon, situation, problem or issue;
- information is gathered using predominantly quantitative variables;
- analysis is geared to ascertain the magnitude of the variation.



STEPS of the RESEARCH PROCESS

